

WASP-177b

R-2439

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-177_230906 · created 2026-08-22T01:39:12+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460193.739 +/- 0.01 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.102 +/- 0.046
Transit depth (Rp/Rs)^2	1.04 +/- 0.95 [%]
Orbital Inclination (inc)	84.92 +/- 2.75
Airmass coefficient 1 (a1)	1.042 +/- 0.016
Airmass coefficient 2 (a2)	-0.025 +/- 0.015
Scatter in the residuals of the lightcurve fit is	1.56 %
Best Comparison Star	#1 - [168, 258]
Optimal Method	PSF photometry
Transit Duration (day)	0.037 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.102 ± 0.046 vs 0.1618 (deviation 1.14σ)
Duration fit vs published	0.037 d vs 0.0721 d (deviation 1.14σ)
Mid-time O–C	22.1 min
Depth SNR	2.0
Residual scatter	1.56 %

Light curve

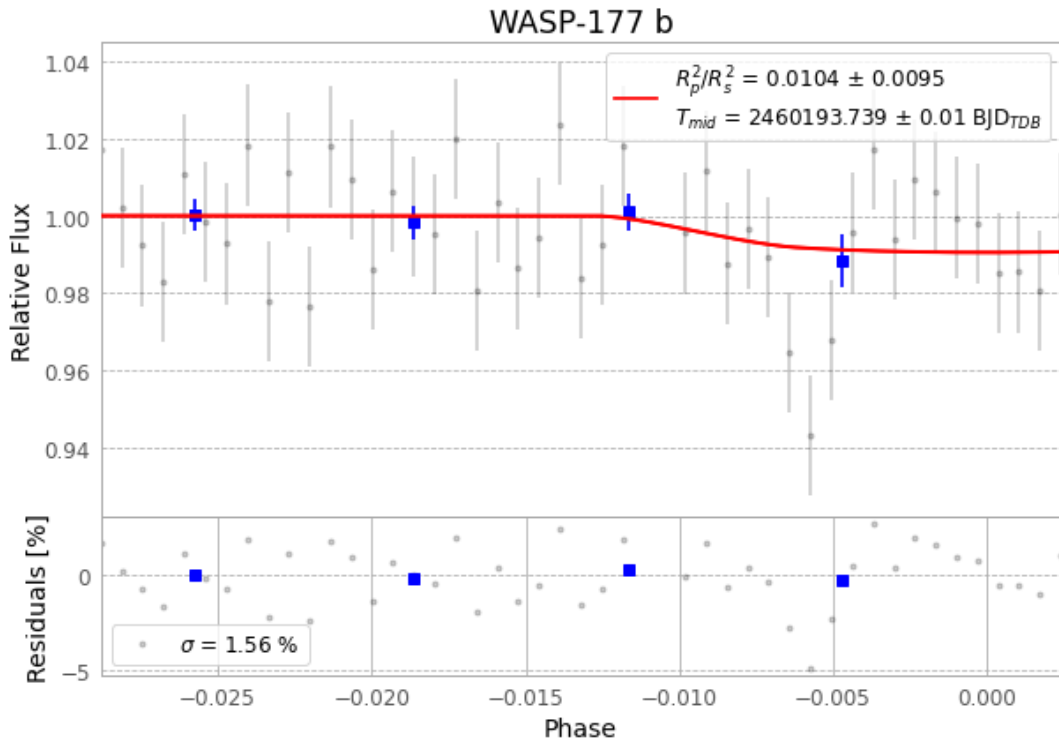


Figure 1 — FinalLightCurve_WASP-177 b_2023-09-05.png (SHA-256 9b230dba16473c5d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [437, 401], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 9baf3917abdcef46...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

47 FITS frames (31 MB), WASP-177230906032715.FITS → WASP-177230906054515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-177-230906.log (49850e05cf78...), FinalLightCurve_WASP-177 b_2023-09-05.png (9b230dba1647...), FinalLightCurve_WASP-177 b_2023-09-05.csv (d4cbd102cbff...)

Compute resources

Wall time 97.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-22 01:39Z.